# Chain Reaction Independent Fission Reactors (CIRs): A Novel Accelerator-Driven Annular Subcritical Architecture Utilizing Intermediate Spallation Neutron Conversion

**Nitinkumar Vishwakarma***
*Department of Electronics and Telecommunication Engineering*
*Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology*
*Andheri, Mumbai, Maharashtra, India*

February 11, 2026

**Abstract**

Traditional nuclear fission power generation is universally founded upon self-sustaining, critical chain reactions, classified herein as Chain Reaction Dependent Reactors (CDRs). Although technologically mature, CDRs enforce severe operational limitations: the requirement for critical mass, delicate reactivity stabilization, costly isotopic enrichment ($^{235}$U), voluminous moderator/reflector assemblies, and inherent vulnerabilities to power excursions or decay-heat loss-of-coolant disasters. This paper presents the conceptual and structural formulation of a novel paradigm: the **Chain Reaction Independent Reactor (CIR)**. Operating in an inherently subcritical regime ($k_{\text{eff}} < 1$), the CIR decouples power generation from autonomous neutron multiplication by utilizing an integrated toroidal particle accelerator to induce fission in abundant, non-fissile fertile isotopes ($^{238}$U, $^{232}$Th).

Motivated by science-fiction depictions of compact, modular energy cores, the CIR is conceived as a comparatively smaller, decentralized power plant capable of supplying reliable, carbon-free electricity to a small town or localized microgrid. Designed for rapid deployment across multiple regions, it provides a safe, economically feasible alternative to replace carbon-heavy coal-fired plants without the threat of runaway meltdowns. We address the fundamental problem of direct charged-particle bombardment—specifically, the strong electrostatic Coulomb repulsion and electronic energy losses in heavy nuclei—by introducing a high-$Z$ intermediate converter mantle (Lead). Accelerated protons impact this mantle to produce uncharged secondary neutrons via spallation. These neutrons experience no electrostatic repulsion, penetrating the fertile $^{238}$U core to induce fast fission and release $\sim 200$ MeV per split. Furthermore, we evaluate two fundamental structural topologies: **Type-1 (Inward-Spiral with Central Core)** and **Type-2 (Outward-Spiral with Peripheral Annular Core)**. We demonstrate that Type-2 is fundamentally superior: it exploits natural centrifugal orbit expansion under a constant magnetic field (enabling permanent magnets) and disperses heat over an expansive outer circumference, lowering surface heat flux and ensuring walk-away passive safety.

**Keywords:** Chain Reaction Independent Reactor (CIR); Subcritical System; Spallation Neutron Conversion; Uranium-238 Fast Fission; Coulomb Barrier Mitigation; Toroidal Accelerator; Annular Core Geometry.

*Department of Electronics and Telecommunication Engineering, Sardar Patel Institute of Technology, Munshi Nagar, Andheri (West), Mumbai 400058, Maharashtra, India.

# 1 Introduction and Paradigm Shift

Global energy infrastructure requires high-density, carbon-free baseload power. Nuclear fission provides an enormous energy density, releasing approximately 200 MeV per fission event compared to only a few electron volts in chemical combustion. However, virtually all civilian nuclear reactors operating today belong to a single physical category: Chain Reaction Dependent Reactors (CDRs).

In a CDR, continuous power production depends entirely upon sustaining an autonomous, self-propagating chain reaction at exact criticality (where exactly one neutron from each fission causes another fission). Operating in this tightly balanced regime introduces major engineering and safety challenges:

- **Critical Mass Limits:** The fissile fuel inventory must exceed a geometric critical mass to prevent excessive neutron leakage. A CDR cannot be made arbitrarily small or thin without the chain reaction dying out.
- **Fuel Scarcity and Enrichment:** Conventional reactors require fissile isotopes, primarily Uranium-235 ($^{235}$U), which makes up only 0.72% of natural uranium. The vast majority (99.27%) is Uranium-238 ($^{238}$U), which cannot sustain a thermal chain reaction and is largely discarded as waste.
- **Bulky Moderators and Reflectors:** High-speed fission neutrons must be slowed down using bulky moderators (water or graphite) and surrounded by thick reflectors, adding immense volume and high-pressure plumbing.
- **Safety Vulnerabilities:** Because the reactor contains a critical mass, any failure of mechanical control rods or loss of coolant can lead to runaway power spikes or post-shutdown meltdowns (as seen at Chernobyl and Fukushima).

To solve these limitations, this paper develops the concept of the **Chain Reaction Independent Reactor (CIR)**. In a CIR, nuclear fission does not rely upon an autonomous chain reaction. Instead, it is driven externally by an integrated particle accelerator bombarding fertile heavy nuclei (such as $^{238}$U) in a subcritical state. When the accelerator beam is turned off, the fission reaction terminates instantly. This removes critical mass limitations, eliminates moderators and reflectors, opens up abundant fertile fuels, and guarantees fail-safe passive safety.

# 2 Conceptual Vision and Motivation: Decentralized Modular Clean Power

The founding motivation of the Chain Reaction Independent Reactor (CIR) draws inspiration from science-fiction depictions of compact, self-contained energy cores—re-imagining nuclear power not as a colossal, multi-billion-dollar industrial complex, but as a sleek, modular, and accessible energy source.

Traditional civilian nuclear stations require vast exclusion zones, heavy containment infrastructure, and continental transmission grids. In contrast, the CIR is conceived as a comparatively small, town-scale power plant engineered to generate reliable electricity for a localized municipality, remote settlement, or industrial microgrid. Its compact physical footprint allows standardized prefabrication, rapid transport, and straightforward deployment across multiple diverse regions directly where power is demanded.

By providing safe, continuous 24/7 baseload energy, the CIR offers a feasible and clean alternative to phase out global dependency on carbon-heavy, polluting coal power plants. Crucially, the architecture prioritizes economic feasibility and fail-safe operation: by eliminating isotopic enrichment, the system burns abundant fertile fuels (natural Uranium-238, Thorium-232, and

depleted uranium stockpiles). Operating subcritically with passive decay-heat dissipation, the reactor completely eliminates the threat of runaway criticality or meltdowns, realizing a safe, decentralized clean-energy future.

# 3 Taxonomic Comparison: CDRs vs. CIRs

The shift from critical CDRs to subcritical CIRs fundamentally alters how a nuclear reactor is constructed and controlled. Table 1 outlines the operational distinctions between the two reactor classes.

Table 1: Operational comparison between Chain Reaction Dependent and Chain Reaction Independent Reactors.

| **Parameter** | **Chain Reaction Dependent (CDR)** | **Chain Reaction Independent (CIR)** |
|---|---|---|
| **Criticality State** | Critical ($k_{\text{eff}} = 1.0$); strictly self-multiplying. | Subcritical ($k_{\text{eff}} < 1.0$); driven by external particle beam. |
| **Primary Nuclear Fuel** | Fissile isotopes ($^{235}$U, $^{239}$Pu); requires enrichment. | Fertile heavy isotopes ($^{238}$U, $^{232}$Th, depleted uranium). |
| **Core Size & Mass** | Constrained by critical mass; minimum size required. | No critical mass limit; size chosen freely for heat removal. |
| **Moderator Assembly** | Compulsory (water, graphite) to slow down neutrons. | Eliminated; fast-spectrum particles drive the reaction. |
| **Neutron Reflectors** | Mandatory to prevent neutron leakage from core. | Unnecessary for reaction; standard shielding is sufficient. |
| **Power Regulation** | Mechanical absorber rods, burnable chemical poisons. | Direct electrical adjustment of the accelerator beam current. |
| **Passive Safety** | Vulnerable to prompt power spikes if controls fail. | Fail-safe; reaction halts instantly when beam cuts out. |
| **Fuel Abundance** | Scarce (only 0.72% of natural uranium). | Abundant (99.27% of natural uranium, 100% of thorium). |
| **Waste Footprint** | Generates long-lived transuranic radioactive waste. | Capable of burning depleted uranium and fertile actinide waste. |

## 3.1 Decoupling of Core Geometry from Critical Mass

In a conventional reactor, the nuclear fuel must exceed a strict minimum physical size called the critical mass. If the core is too small or too thin, too many neutrons escape through the outer boundary before they can strike another nucleus, causing the chain reaction to immediately extinguish. This imposes rigid geometric constraints: conventional reactor cores must always be thick, bulky, and compact cylinders.

In a CIR, because the reactor does not rely on a self-sustaining chain reaction, neutron leakage does not kill the energy generation process. Power is driven purely by the incoming external particle beam. As a result, the fuel core is completely freed from critical mass limits. The core can be made significantly smaller, divided into modules, or arranged into a thin circular ring designed purely for optimal heat extraction.

## 3.2 Elimination of Moderators and Reflectors

In conventional reactors, neutrons born from fission travel at very high speeds (fast neutrons). However, Uranium-235 splits most easily when struck by very slow, thermal neutrons. Therefore, traditional reactors must pack the core with massive volumes of moderators (such as light water, heavy water, or graphite) to slow the neutrons down, and surround the core with heavy reflectors to bounce escaping neutrons back inside.

In a CIR, the fertile fuel (Uranium-238) is fissioned by fast-moving particles and fast neutrons. Slowing the neutrons down is completely unnecessary. By discarding bulky moderator tanks, high-pressure moderator pipes, and heavy reflectors, the physical footprint, weight, and complexity of the reactor are drastically reduced.

### 3.3 Fuel Abundance: Direct Utilization of Uranium-238

Natural uranium is composed of approximately 99.27% Uranium-238 and only 0.72% Uranium-235. Because conventional reactors cannot burn Uranium-238 directly, they require multi-million-dollar enrichment facilities, leaving behind huge stockpiles of depleted uranium as useless radioactive waste. Furthermore, Thorium-232—which is three to four times more abundant than uranium on Earth—is entirely fertile and non-fissile.

Because CIRs do not require self-sustaining fissile reactions, they can use Uranium-238 and Thorium-232 directly as primary fuel. This unlocks centuries of energy from existing depleted uranium stockpiles without needing any enrichment plants.

### 3.4 Inherent Deterministic Passive Safety

In a conventional reactor, the core contains enough fissile material to explode in power if control systems fail. The plant must constantly balance the reaction using mechanical control rods. If control rods jam or an emergency occurs, a dangerous power excursion or core meltdown can happen.

In a CIR, the core is subcritical ($k_{\text{eff}} < 1$). The fuel by itself is physically incapable of sustaining a chain reaction. Fission only occurs while the external particle beam is striking the fuel. If an earthquake, electrical blackout, or equipment failure occurs, the accelerator beam simply turns off. The moment the beam shuts off, all fission reactions stop completely in less than a microsecond. The reactor is fail-safe by the laws of physics.

## 4 Accelerator Design in Toroidal Geometry

The driving unit of the CIR is an integrated particle accelerator housed inside a circular vacuum tube arranged like a hollow torus. Charged particles (such as protons from hydrogen gas) are fed into the vacuum chamber and accelerated along circular paths.

### 4.1 Particle Kinematics and Orbit Radius

When a charged particle of mass $m$ and charge $q$ moves with velocity $v$ through a perpendicular magnetic field $B$, the magnetic force bends the particle into a circular path of radius $r$, described by the cyclotron relation:

$$r = \frac{m \cdot v}{q \cdot B} \tag{1}$$

This formula reveals the fundamental rule of the accelerator: as the particle gains speed ($v$ increases), its orbital radius ($r$) naturally wants to grow, causing the particle to trace an expanding circular path unless contained by the magnetic field.

### 4.2 Azimuthal Acceleration Coils (Coil-2 Array)

To increase the speed and kinetic energy of the particles, an arrangement of electromagnetic coils (designated as "Coil-2") is distributed along the length of the circular tube. These coils turn on and off in a systematic, high-frequency rhythm. As the coils oscillate between positive and negative polarities, the alternating electric force pulls and pushes the charged particle forward at every station. With each complete revolution around the torus, the particle receives repeated kinetic energy boosts until it reaches the desired velocity.

### 4.3 Magnetic Guidance Coils (Coil-1 Array)

To keep the particles safely inside the circular vacuum channel as they accelerate, a primary magnetic field is generated by guiding coils (designated as "Coil-1"). This magnetic field provides the centripetal force directed toward the center, preventing the particles from flying into the walls before they reach the target.

## 5 Structural Topologies: Type-1 vs. Type-2 CIR

Depending on where the fuel core and feeders are placed, CIRs can be divided into two main configurations: Type-1 and Type-2.

### 5.1 Type-1 CIR: Inward-Spiral Configuration

In the Type-1 design, the nuclear fuel core and heat extraction unit are placed at the center of the torus. The feeders inject particles at the outer edge, and the particles are forced to spiral inward toward the central core as they accelerate.

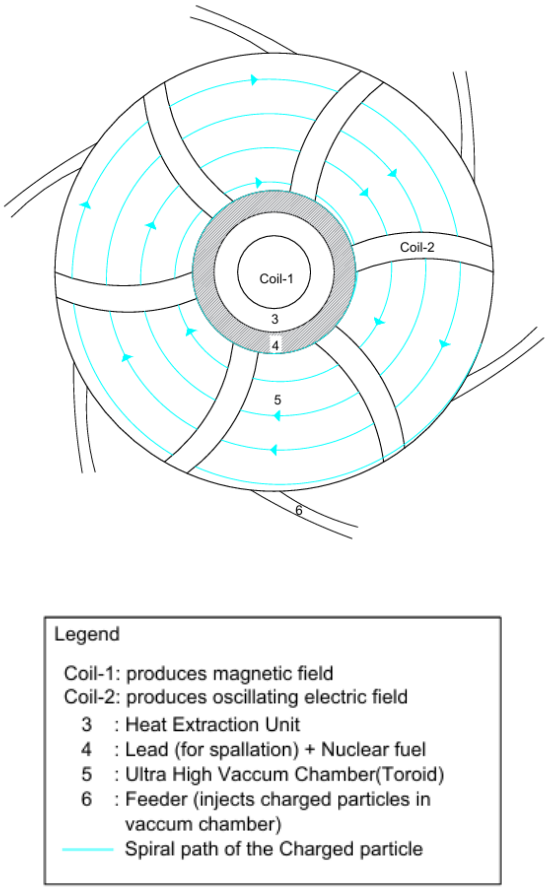



Figure 1: Schematic representation of the Type-1 CIR inward-spiral system with engineering component legend.

### 5.1.1 Electrodynamic Limitations of Type-1 Systems

The Type-1 design suffers from a severe physical disadvantage. As particles gain speed, natural centrifugal force wants to push them outward to a larger radius. However, to make them hit a central core, the system must force them inward into an increasingly tighter circle while they are moving faster and faster. To do this, the magnetic coils at the center must generate extremely intense, rapidly pulsing magnetic fields that constantly fight against centrifugal force. This consumes an enormous amount of electrical power just to bend the beam inward. Furthermore, placing the core in the center concentrates all the fission heat into a tiny space, making it very difficult to cool.

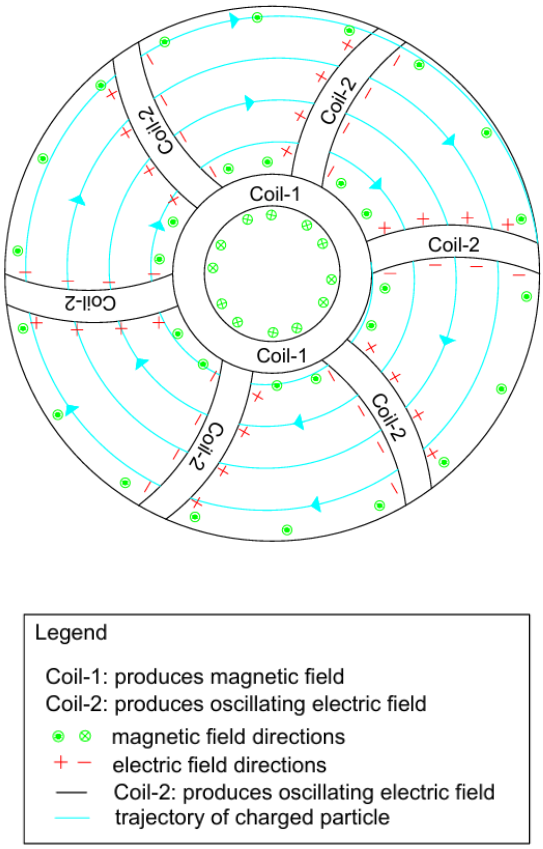



Figure 2: Coil arrangements and field vectors for toroidal acceleration with electromagnetic orientation legend.

## 5.2 Type-2 CIR: Outward-Spiral Configuration

In the Type-2 design, the feeders inject particles near the center of the torus, and the nuclear fuel core is placed along the outer circumference of the torus. As the particles accelerate, their radius naturally expands outward in a smooth spiral until they reach the predetermined speed and strike the outer core.

### 5.2.1 Decisive Advantages of the Type-2 Design

- **Constant Magnetic Field (Permanent Magnets):** Because the particle naturally spirals outward as its velocity increases ($r = mv/qB$), the magnetic field inside the vacuum

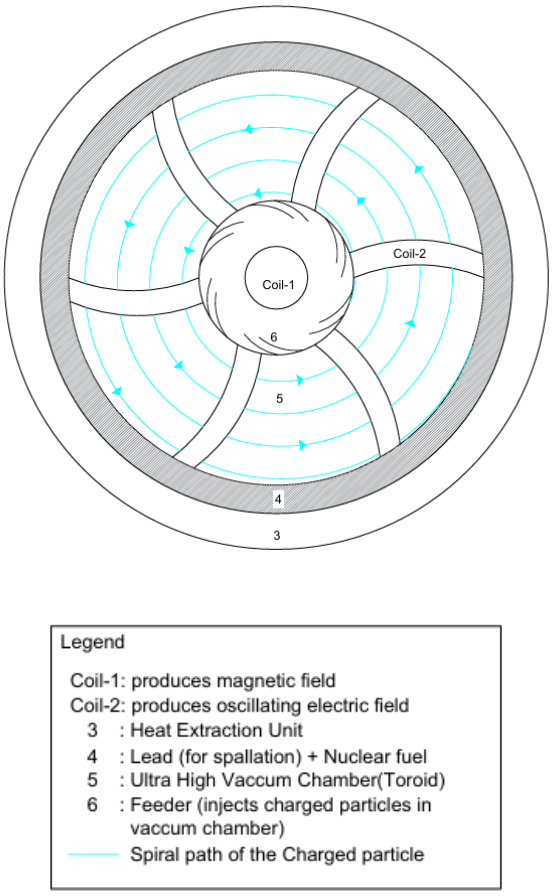



Figure 3: Structural architecture of the Type-2 CIR outward-spiral system with engineering component legend.

chamber can be kept completely constant. This means the guiding field can even be produced using permanent magnets, requiring zero electrical power for beam bending.

- **Larger Surface Area for Cooling:** In Type-2, the nuclear core forms a wide circular ring around the outside of the reactor. This gives the core a vastly larger surface area, spreading the heat out over a wide region and making heat extraction far more efficient.
- **Lower Peak Core Temperatures:** Because the heat is spread over a wide outer mantle rather than concentrated in a tiny central dot, the core temperature stays at manageable levels, drastically reducing wasted radiant heat to the surroundings.

Table 2: Comparison between Type-1 and Type-2 CIR architectures.

| **Engineering Feature** | **Type-1 (Inward Spiral)** | **Type-2 (Outward Spiral)** |
|---|---|---|
| **Particle Orbit** | Inward spiral (outer edge toward center). | Outward spiral (center toward outer ring). |
| **Magnetic Field** | Rapidly pulsing, high-power magnetic field required. | Constant magnetic field; permanent magnets can be used. |
| **Magnet Energy Consumption** | High (fights against natural centrifugal force). | Zero to minimal (works with natural orbit expansion). |
| **Nuclear Core Location** | Compact central core at the middle of torus. | Expansive circular ring on the outer perimeter. |
| **Cooling Surface Area** | Very small (tight central cylinder). | Very large (outer circumference of the reactor). |
| **Heat Extraction** | Difficult; high localized heat concentration. | Easy and uniform; heat is distributed widely. |
| **Target Impact** | Requires complex steering to force beam inward. | Natural impact when particle reaches outer ring radius. |

# 6 Overcoming Coulomb Repulsion: The Intermediate Spallation Layer

## 6.1 The Coulomb Repulsion Problem

In the original CIR concept, accelerated charged particles (such as protons) are smashed directly into the heavy Uranium-238 nucleus. However, this faces a major physics problem: electrostatic Coulomb repulsion.

A proton has a positive charge (+1), and a Uranium nucleus has 92 positive protons (+92). Because like charges repel, the uranium nucleus strongly repels any incoming proton as it gets close. The electrostatic repulsion energy between the proton and uranium nucleus is given by:

$$V_C = \frac{1}{4\pi\varepsilon_0}\frac{Z_1 \cdot Z_2 \cdot e^2}{R} \approx 14.7 \text{ MeV} \tag{2}$$

This means a proton needs approximately 14.7 MeV of kinetic energy just to overcome the electrostatic repulsion and touch the uranium nucleus.

### 6.1.1 Why Direct Proton Bombardment Fails

One might assume we could simply accelerate protons to high kinetic energies (several MeV) to overpower the repulsion. However, uranium metal is filled with a dense cloud of atomic electrons. When an energetic proton enters solid uranium, it collides millions of times with surrounding electrons long before it ever encounters a tiny atomic nucleus. Nearly all of the proton's kinetic energy is stripped away through atomic ionization, turning into useless low-grade heat like an

electric heating element. As a result, the accelerator would consume far more electrical power than the occasional nuclear fission could ever produce, making the reactor an energy-losing device.

### 6.2 The Intermediate Converter Solution: The Lead Spallation Mantle

To solve this problem, an intermediate layer of heavy metal (specifically Lead) is placed on the inside edge of the outer nuclear core ring (as shown in Figure 3, component 4). The reaction sequence operates in three distinct stages:

1. **Stage 1 (Proton Impact on Lead):** Accelerated protons strike the outer lead layer first instead of hitting uranium directly.
2. **Stage 2 (Neutron Production):** When a high-energy proton hits a heavy lead nucleus, it shatters the nucleus through spallation, freeing a shower of fast secondary neutrons.
3. **Stage 3 (Zero Repulsion Fission):** Crucially, neutrons carry ZERO electrical charge (neutral). Because a neutron has no charge, it experiences ZERO Coulomb repulsion from the uranium nucleus.

### 6.3 Fast-Neutron Fission in Uranium-238

Because neutrons experience no electrostatic repulsion, they pass easily through the lead layer and penetrate deep into the Uranium-238 fuel. When an energetic neutron (energy $> 1.1$ MeV) strikes a Uranium-238 nucleus, it causes fast fission, splitting the uranium atom and releasing approximately 200 MeV of heat energy per split.

Furthermore, a single energetic proton hitting the lead layer produces multiple secondary neutrons. Each of these neutrons can cause a separate fission in a different uranium atom. This multi-neutron multiplier effect ensures that the nuclear energy released is vastly greater than the electrical energy used to accelerate the original proton, achieving true net energy gain.

## 7 Thermal Hydraulics and Walk-Away Passive Safety

### 7.1 Heat Extraction Area in Annular Geometry

In a conventional reactor, all fuel is packed into a tight cylinder of small radius $R_{\text{in}}$. In the Type-2 CIR, the core is a wide ring of outer radius $R_{\text{out}}$. The heat transfer surface area of a cylinder of height $H$ is:

$$A = 2 \cdot \pi \cdot R \cdot H \tag{3}$$

Because $R_{\text{out}}$ is many times larger than $R_{\text{in}}$, the cooling surface area of the Type-2 core is roughly 5 to 10 times larger than a conventional reactor core. The surface heat flux (heat per square meter) is:

$$\text{Heat Flux} = \frac{\text{Thermal Power}}{2 \cdot \pi \cdot R_{\text{out}} \cdot H} \tag{4}$$

This drastically reduces the thermal strain on fuel elements, prevents hot spots, and allows heat to be carried away smoothly by cooling fluids (such as liquid lead-bismuth, helium gas, or molten salt) to power an electrical turbine.

### 7.2 Passive Decay-Heat Removal (Walk-Away Safety)

In a conventional nuclear reactor, even after the reactor is turned off, unstable fission products continue to decay radioactively, generating "decay heat" (about 7% of full power immediately after shutdown). In a conventional dense core, this decay heat is so concentrated that if water

pumps lose electricity (as occurred during the Fukushima disaster), the water boils dry and the fuel rods melt into radioactive lava.

In the Type-2 CIR, because the fuel is distributed across a wide, thin outer ring with a huge surface area, the decay heat per square meter is extremely small. Even in a total electrical blackout with all cooling pumps dead, natural air circulation and heat radiation from the outer mantle are sufficient to carry away the decay heat completely. The fuel will never melt. The reactor possesses true "walk-away" passive safety.

## 8 Practical Engineering Considerations

- **Vacuum Maintenance:** The toroidal chamber must maintain a high vacuum so accelerated particles do not collide with air molecules during flight.
- **Liquid Metal Target:** Using liquid Lead-Bismuth Eutectic (LBE) in the converter layer provides an ideal target because a liquid cannot suffer solid radiation damage or cracking, and it simultaneously acts as an excellent heat transfer fluid.
- **Permanent Magnet Integration:** In the Type-2 design, permanent magnet arrays can provide the guiding magnetic field, eliminating the electrical cost of running continuous magnet coils.

## 9 Conclusion

The Chain Reaction Independent Reactor (CIR) represents a fundamental departure from century-old critical reactors. By abandoning the constraint of a self-sustaining chain reaction in favor of an externally driven subcritical core, the CIR eliminates critical mass limitations, dispenses with bulky moderators and reflectors, and unlocks the vast global reserves of non-enriched Uranium-238.

The electrostatic Coulomb barrier is solved through a bilayer target featuring an intermediate lead spallation mantle, which converts accelerated protons into uncharged fast neutrons that penetrate Uranium-238 without repulsion. Furthermore, the outward-spiral Type-2 architecture is fundamentally superior: it operates under a constant magnetic field (allowing permanent magnets), works in harmony with natural orbital expansion, and spreads fission heat over an expansive outer ring for guaranteed walk-away passive safety.

## References


[1] C. Rubbia, et al., "Conceptual design of a fast neutron operated high power energy amplifier," European Organization for Nuclear Research (CERN), Geneva, Report CERN/AT/95-44, 1995.

[2] C. D. Bowman, et al., "Nuclear energy generation and waste transmutation using an accelerator-driven intense thermal neutron source," *Nuclear Instruments and Methods in Physics Research Section A*, vol. 320, pp. 336–367, 1992.

[3] H. Aït Abderrahim, et al., "MYRRHA: A multipurpose hybrid research reactor for high-tech applications," *Nuclear Physics News*, vol. 20, no. 1, pp. 24–28, 2010.

[4] J. J. Duderstadt and L. J. Hamilton, *Nuclear Reactor Analysis*, John Wiley & Sons, New York, 1976.

[5] E. O. Lawrence and M. S. Livingston, "The production of high speed light ions without the use of high voltages," *Physical Review*, vol. 40, no. 1, pp. 19–35, 1932.

[6] S. Humphries, *Principles of Charged Particle Acceleration*, John Wiley & Sons, New York, 1999.

[7] J. R. Lamarsh and A. J. Baratta, *Introduction to Nuclear Engineering*, Prentice Hall, Upper Saddle River, NJ, 2001.